

Thermodynamics of Metal–Acetate Interactions

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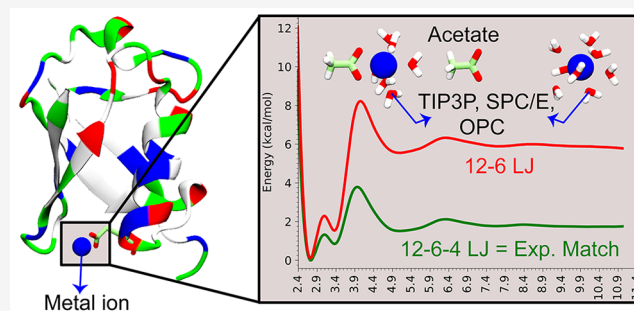
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ABSTRACT: Metal ions play crucial roles in protein- and ligand-mediated interactions. They not only act as catalysts to facilitate biological processes but are also important as protein structural elements. Accurately predicting metal ion interactions in computational studies has always been a challenge, and various methods have been suggested to improve these interactions. One such method is the 12–6–4 Lennard-Jones (LJ)-type nonbonded model. Using this model, it has been possible to successfully reproduce the experimental properties of metal ions in aqueous solution. The model includes induced dipole interactions typically ignored in the standard 12–6 LJ nonbonded model. In this we expand the applicability of this model to metal ion–carboxylate interactions. Using 12–6–4 parameters that reproduce the solvation free energies of the metal ions leads to an overestimation of metal ion–acetate interactions, thus, prompting us to fine-tune the model to specifically handle the latter. We also show that the standard 12–6 LJ model significantly falls short in reproducing the experimental binding free energy between acetate and 11 metal ions (Ni(II), Mg(II), Cu(II), Zn(II), Co(II), Cu(I), Fe(II), Mn(II), Cd(II), Ca(II), and Ag(I)). In this study, we describe optimized C_4 parameters for the 12–6–4 LJ nonbonded model to be used with three widely employed water models (Transferable Intermolecular Potential with 3 Points (TIP3P), Simple Point Charge Extended (SPC/E), and Optimal Point Charge (OPC) water models). These parameters can accurately match the experimental binding free energy between 11 metal ions and acetate. These parameters can be applied to the study of metalloproteins and transition metal ion channels and transporters, as acetate serves as a representative of the negatively charged amino acid side chains from aspartate and glutamate.



1. INTRODUCTION

Metal ions have diverse roles in protein-mediated processes like cell signaling pathways and DNA synthesis.¹ These metal ions contribute to protein activity by facilitating the transfer of electrons between atoms and even serving as structural components and catalytic cofactors.^{1–3} Metal ions facilitate electron transfer over relatively long molecular distances in enzymes such as ribonucleotide reductase⁴ and photosystem II.⁵ Moreover, metal ions can trigger protein structure changes in metalloregulatory proteins including NikR and Fur protein families.³ Metal ions are also transported across plasma membranes of eukaryotic or prokaryotic cells by metal ion channels and transporters, either directly or in complex with metal chelators like siderophores.^{3,6,7} Therefore, having models that accurately represent metal ion interactions with protein environments is essential to decipher the structure and function of metalloproteins and metalloenzymes as well as metal ion channels and transporters that are membrane proteins.³

The accurate description of the interaction between metal ions and proteins at the atomic level still represents a challenge

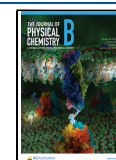
for both experiments¹ and simulations. Several computational methods have been developed to compute the interactions between metal ions and their surrounding environment.⁸ However, none of these methods are flawless, and each has its own limitations. In the context of force-field-based molecular simulations, our group has developed the 12–6–4 LJ nonbonded model, a model that incorporates a $1/r^4$ term into the standard 12–6 LJ potential.⁸ This addition accommodates ion-induced dipole interactions as C_4 terms, which previously were overlooked. Li and Merz⁸ determined that the standard 12–6 LJ nonbonded model cannot accurately reproduce certain experimental properties, such as metal ion coordination numbers, ion–oxygen distances, and hydration free energies. In light of this observation, they

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developed the novel 12–6–4 LJ model, which is capable of reproducing these experimental properties within an acceptable margin of error.

The 12–6–4 LJ model was initially established to compute the interactions between metal ions and water molecules. However, in metalloproteins and metalloenzymes, the metal ion coordinates with additional atoms. Metal ions in proteins and enzymes are mostly coordinated by the oxygen, nitrogen, and sulfur atoms, which are provided by water molecules and amino acid side chains.⁹ Li, Merz, and their co-workers¹⁰ discovered that both the 12–6 LJ and 12–6–4 LJ nonbonded models fail to accurately reproduce the experimental free energies of metal ions with ligands, including the histidine residue. As a result, they introduced new parameters to fine-tune the C_4 terms in the 12–6–4 LJ model for 11 metal ions that interact with imidazole, this molecule representing the histidine residue side chain. They demonstrated that by optimizing the polarizability of the chelating atom of imidazole (N atom), they were able to accurately replicate the experimental binding free energies between imidazole and metal ions, whereas the standard 12–6 LJ model failed to do so. This discovery serves as a further validation of the capabilities of the newly developed 12–6–4 LJ nonbonded model.¹⁰

A significant number of metalloproteins are available in the Protein Data Bank (PDB)^{11,12} with at least one histidine residue coordinating a metal ion.¹⁰ The same fact holds true for acidic residues (i.e., aspartate and glutamate). Indeed, negatively charged residues are also essential for the function and stability of metalloenzymes and metalloproteins.^{13,14} However, there is a lack of parameters available for negatively charged residues while chelating to metal ions within the 12–6–4 LJ nonbonded model. Here, we used acetate (CH_3COO^-) as a minimal molecular model representing the side chains of these residues. Our results demonstrate that the standard 12–6 LJ and 12–6–4 LJ models cannot reproduce the experimental interaction energies of 11 metal ions with the acetate molecule. In the present study, we provide the adjusted C_4 parameters for the 12–6–4 LJ nonbonded model that can be used to accurately model the interactions between the COO group and various metal ions in aqueous solutions using three different water models (namely, TIP3P, SPC/E, and OPC).

2. METHODS

2.1. Refining the C_4 Terms in the 12–6–4 LJ-Type Nonbonded Model. Due to the popularity of nonbonded models and the recent advances in the accuracy of the 12–6–4 model, we conducted a fine-tuning of the C_4 term parameters specifically for acidic residues and metal ions. With that in mind, to achieve the experimental binding free energies for the complex of a metal ion with an acetate molecule, we modified the chelator atom polarizability, specifically the polarizability of the acetate oxygen atoms (termed α_0 in the following equation, the angle generated by the metal ion between the induced dipole and electronic field is defined by θ):

$$C_4 \approx -\frac{1}{2}\alpha_0\left(\frac{q}{4\pi\epsilon_0\epsilon_r}\right)^2 \cos \theta$$

The following equation represents the 12–6–4 LJ model in the AMBER force field.^{8,15,16} In this equation, r represents the distance between particles i and j . The $-\frac{C_{12}^{ij}}{r_{ij}^{12}}$ part represents the

C_4 term, which describes the ion–induced dipole interaction and is inversely proportional to r^4 . The final part of the equation represents the Coulomb pair potential, where e represents the charge of a proton and Q_i and Q_j denote the partial charges of atom i and j , respectively:

$$U_{ij}^-(r_{ij}) = \frac{C_{12}^{ij}}{r_{ij}^{12}} - \frac{C_6^{ij}}{r_{ij}^6} - \frac{C_4^{ij}}{r_{ij}^4} + \frac{e^2 Q_i Q_j}{r_{ij}}$$

$$= 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] - \frac{C_4^{ij}}{r_{ij}^4} + \frac{e^2 Q_i Q_j}{r_{ij}}$$

Initially, we constructed a topology file by employing classical 12–6 Lennard-Jones (LJ) nonbonded parameters. We then used this topology file to generate a new one, incorporating the default C_4 terms. Because no optimized C_4 parameters were available for the acetate–metal ion interaction, we generated a revised topology file using select polarizability values for the acetate target atoms. We conducted an unbiased molecular dynamics (MD) simulation using the newly generated topology file to determine the equilibrium distance between acetate and the metal ion. This equilibrium distance was the primary distance for undertaking Umbrella Sampling (US) simulations. The US method, coupled with the weighted histogram analysis algorithm (WHAM),¹⁷ has been widely used in computational studies to determine the potential of mean force (PMF) energy of metal ions in various environments.^{10,18–22} We systematically examined various C_4 values and compared the calculated interaction energies with the experimental interaction energies as our benchmark.

2.2. Potential of Mean Force and Molecular Dynamics Simulations. The PMF obtains the free energy profile of a system with respect to generalized coordinates such as Ramachandran angles or the distance between two atoms. The PMF free energy profile serves as a visual representation that reveals the stable conformations of a given molecule as well as the barriers the molecule needs to overcome when transitioning from one state to another along a defined reaction coordinate.²³

To assess the convergence of PMF calculations, we initially performed the US method using a 2 ns simulation length for each window. Subsequently, upon obtaining the experimental binding free energy, we conducted the US method for a given metal ion with simulation lengths of 2, 4, 6, 8, and 10 ns for each window. While we opted for a 4 ns simulation length for each window in SPC/E water,²⁴ we discovered that the optimal simulation length for each window in TIP3P²⁵ and OPC²⁶ water is 6 ns. The reaction coordinate involved the distance between the carbon atom of the acetate carboxyl group and the metal ion. The sampling window width ranged from 0.05 Å (<5 Å) to 0.1 Å (>5 Å), depending on the metal ion's proximity to acetate. The starting point for sampling ranged from 1.8 to 2.5 Å, determined by the metal ion's radius.

To assess the precision of our computed binding free energies, three replicas of PMF calculations were performed. The replicas showed an acceptable margin of error of less than 0.35 kcal/mol. To further validate the obtained results using the US algorithm as implemented within the AMBER software package, we also performed additional PMF calculations using the US method as implemented in PLUMED software.^{27–29} The latter binding free energies were consistent with those acquired from the standard implementation in AMBER. Note

that in the following, we will refer to either AMBER or PLUMED PMF data according to the procedure described above.

All the MD simulations were performed using the AMBER 20 package³⁰ and the PLUMED package version 2.8.2.^{27–29} A four-step method of minimization was employed to minimize the system's energy. Only water molecules were minimized in the first step, while the acetate molecule and metal ion were restrained. In the second step, another minimization was performed on the acetate hydrogen atoms and water molecules. The third step involved a minimization of acetate non-hydrogen atoms and water molecules. Finally, in the last step, a minimization was performed on the entire system. In each stage, 10000 steps of the steepest descent were used. Afterward, the system was gradually heated from 0 to 300 K over 1 ns in the NVT ensemble. The heated system was then used for a 4 ns simulation in the NPT ensemble. The Berendsen barostat,³¹ employing a pressure relaxation time of 5 ps, was applied to maintain the system's pressure. The Langevin thermostat was used as the temperature coupling algorithm. To constrain bond lengths, the SHAKE algorithm³² was employed, and periodic boundary conditions were applied in all three dimensions (*x*, *y*, and *z*). A cutoff distance of 10 Å was applied to handle nonbonded interactions. The data analysis and visualization in the present work were performed using the Visual Molecular Dynamics (VMD) and Python packages.^{33,34}

2.3. MD Simulation of the Bacterial Glyoxalase I (Glx I) Metalloprotein. The initial protein structure was obtained from the PDB (PDB ID 1F9Z).³⁵ This structure was used to generate a simulation system comprising the OPC water model and the metalloprotein. A five-stage minimization process was conducted, with 10000 steps of steepest descent employed in each stage to optimize the simulation system. Afterward, a 5 ns NVT ensemble was performed to gradually raise the temperature from 0 to 300 K. Following this, the system underwent equilibration for 5 ns by using the NPT ensemble. The equilibrated system was followed by a 200 ns MD simulation using AMBER 20. The SHAKE algorithm³² was applied to constrain all bonds, while the Langevin dynamics algorithm was used to maintain the temperature at 300 K. Isotropic pressure scaling with the Brendsen barostat³¹ was utilized to control pressure.

3. RESULTS AND DISCUSSION

In recent years, researchers have dedicated efforts to developing and optimizing polarizable force fields for transition metal ions.^{36–38} In some instances, they successfully reproduced experimental and ab initio calculation results. Despite polarizable force fields being more complicated and accurate than fixed-charge force fields, there are cases where the results from polarizable force fields are not as good as those from fixed-charge force fields.^{39–41} One potential reason for this discrepancy could be the challenging nature of optimizing polarizable force fields.⁴¹

A recent investigation into the binding of acetate to divalent metal ions, such as Mg(II), Ca(II), and Zn(II), indicates that the AMOEBA (Atomic Multipole Optimized Energetics for Biomolecular Applications) force field⁴² has the potential to enhance the prediction of binding constants for acetate–cation complexes. However, it appears to overestimate the binding strength of the metal ions, leading to results that do not align quantitatively with experimental data.⁴³ Jing et al.⁴⁴ used the

AMOEBA force field to investigate the interactions between metal ions and protein compounds. The results showed that although AMOEBA predicts the solvation free energy of acetate within a satisfactory range, some discrepancies are noted for acetate–metal ion interaction energies, particularly for the polarization energy at short intermolecular distances.

Researchers also evaluated the variation of interaction energies of different protein zinc binding sites⁴⁵ and calcium acetate⁴⁶ using QM calculations as the benchmark to evaluate polarizable force field accuracy. Their results showed that the CHARMM-Drude polarizable model^{47–49} agrees with the QM benchmark, with errors around 10–80 kcal/mol; however, there were some remaining discrepancies.

In the current work, we applied the AMBER ff14SB nonpolarizable force field to predict the binding free energy of acetate with 11 metal ions. Our results indicated that the reparametrized 12–6–4 LJ model can reproduce the experimental interaction energy of the acetate–metal ion complexes within a margin of error of less than 0.35 kcal/mol.

3.1. Effect of Water Model and Polarizability on Binding Free Energy Calculations. Table 1 gives the experimental free energies of association between 11 divalent metal ions and acetate along with their electronic configuration and ionic radii.^{50–52}

Table 1. Experimental Binding Free Energy (Exp ΔG) of Each Metal Ion with Acetate, the Ion Electron Configuration (elec conf), and the Ion Radius (*r*) in picometers

ion	exp ΔG (kcal/mol) ^{50–52}	elec conf	<i>r</i> (pm) ⁵³
Ni(II)	−1.95	[Ar] 3d ⁸	69
Mg(II)	−1.33	[Ne]	72
Cu(II)	−3.01	[Ar] 3d ⁹	73
Zn(II)	−2.16	[Ar] 3d ¹⁰	74
Co(II)	−1.88	[Ar] 3d ⁷	75
Cu(I)	−2.78	[Ar] 3d ¹⁰	77
Fe(II)	−1.91	[Ar] 3d ⁶	78
Mn(II)	−1.91	[Ar] 3d ⁵	83
Cd(II)	−2.63	[Kr] 4d ¹⁰	95
Ca(II)	−1.27	[Ar]	100
Ag(I)	−1.00	[Kr] 4d ¹⁰	115

Overall, there appears to be no consistent trend between the polarizability value and binding free energy. This phenomenon is also observed with metal ion–imidazole complexes,¹⁰ implying that other interactions, such as water–water interactions surrounding the ions, play a significant role in the binding free energy. It is important to note that the selection of water molecules employed in simulations also has a crucial influence on the interaction free energy. Tables 2 and 3 give the computed free energies between the 11 metal ions and acetate in the three-point TIP3P and SPC/E water models, respectively, and the results for the four-point OPC water model are given in Table 4.

Interestingly, our findings indicate that reproducing the experimental interaction energy requires employing lower polarizability values for the chelator atom within the OPC water model compared to the TIP3P and SPC/E water models (compare Tables 2 and 3, with Table 4).

The OPC water model was designed for improved accuracy in mimicking the properties of liquid water compared with TIP3P and SPC/E, which was achieved through the

Table 2. Calculated Binding Free Energy of Each Metal Ion–Carboxylate Complex in TIP3P Water; Rep #1, Rep #2, and Rep #3 Show the Results from All Three Replicas Performed Using AMBER; Pol and C₄ Are the Applied Polarizability and C₄ Terms, Respectively^a

TIP3P		computed ΔG									
ion	exp ΔG^{50-52}	AMBER					12-6-4 LJ	12-6 LJ	PLUMED	pol ($\text{\AA}^3$)	C_4 (kcal/(mol $\text{\AA}^4$))
		Rep #1	Rep #2	Rep #3	av $\pm$ std						
Ni(II)	-1.95	-2.07	-1.69	-2.00	-1.92 ± 0.2	-1.19	-8.15	-1.89	0.644	94.91	
Mg(II)	-1.33	-1.28	-1.16	-1.30	-1.25 ± 0.07	-3.91	-8.74	-1.42	0.125	11.50	
Cu(II)	-3.01	-3.21	-3.20	-3.26	-3.22 ± 0.03	0.24	-7.75	-3.17	0.944	190.17	
Zn(II)	-2.16	-2.21	-2.74	-2.01	-2.32 ± 0.38	-0.75	-8.31	-2.24	0.730	117.08	
Co(II)	-1.88	-1.95	-1.93	-1.81	-1.90 ± 0.08	-1.15	-7.86	-1.74	0.657	95.41	
Cu(I)	-2.78	-2.61	-2.88	-2.89	-2.79 ± 0.16	-1.86	-2.65	-2.67	1.700	8.24	
Fe(II)	-1.91	-2.08	-1.84	-1.82	-1.91 ± 0.14	-2.87	-7.14	-1.94	0.432	48.76	
Mn(II)	-1.91	-2.06	-1.90	-1.61	-1.86 ± 0.23	-2.57	-7.35	-1.96	0.469	47.45	
Cd(II)	-2.63	-2.53	-2.67	-2.76	-2.65 ± 0.12	0.01	-5.88	-2.79	0.894	114.91	
Ca(II)	-1.27	-1.25	-1.31	-1.00	-1.19 ± 0.16	-1.22	-3.48	-1.16	0.600	36.27	
Ag(I)	-1.00	-1.05	-1.03	-0.97	-1.02 ± 0.04	1.08	-0.13	-1.00	1.500	86.22	

^aThe free energy values are given in kcal/mol.

Table 3. Calculated Binding Free Energy of Each Metal Ion–Carboxylate Complex in SPC/E Water; Rep #1, Rep #2, and Rep #3 Show the Results from All Three Replicas Performed Using AMBER; Pol and C₄ Are the Applied Polarizability and C₄ Terms, Respectively^a

SPC/E		computed ΔG									
ion	exp ΔG^{50-52}	AMBER					12-6-4 LJ	12-6 LJ	PLUMED	pol ($\text{\AA}^3$)	C_4 (kcal/(mol $\text{\AA}^4$))
		Rep #1	Rep #2	Rep #3	av $\pm$ std						
Ni(II)	-1.95	-2.28	-1.82	-2.26	-2.12 ± 0.26	-5.12	-7.50	-1.97	0.450	63.95	
Mg(II)	-1.33	-1.42	-1.29	-1.75	-1.49 ± 0.24	-5.71	-5.39	-1.10	0.125	10.58	
Cu(II)	-3.01	-3.28	-2.82	-3.22	-3.11 ± 0.25	0.11	-6.14	-3.10	1.069	225.35	
Zn(II)	-2.16	-2.37	-1.94	-1.80	-2.04 ± 0.30	-0.20	-6.41	-1.64	0.869	139.14	
Co(II)	-1.88	-2.00	-2.08	-1.90	-1.99 ± 0.09	-0.31	-6.98	-2.40	0.819	118.65	
Cu(I)	-2.78	-3.04	-3.02	-3.09	-3.05 ± 0.04	-1.28	-1.96	-2.70	2.500	15.58	
Fe(II)	-1.91	-1.84	-2.19	-1.83	-1.95 ± 0.21	-1.67	-6.51	-2.21	0.657	201.10	
Mn(II)	-1.91	-2.31	-1.97	-2.01	-2.10 ± 0.19	-1.23	-5.37	-2.00	0.694	74.45	
Cd(II)	-2.63	-2.88	-2.84	-2.32	-2.68 ± 0.31	0.44	-4.15	-2.59	1.000	137.67	
Ca(II)	-1.27	-1.28	-1.50	-1.47	-1.42 ± 0.12	-0.61	-2.55	-1.50	0.869	53.56	
Ag(I)	-1.00	-0.97	-1.09	-1.29	-1.12 ± 0.16	0.74	0.41	-1.08	1.550	98.75	

^aThe free energy values are in kcal/mol.

Table 4. Calculated Binding Free Energy of Each Metal Ion–Carboxylate Complex in OPC Water; Rep #1, Rep #2, and Rep #3 Show the Results from All Three Replicas Performed Using AMBER; Pol and C₄ Are the Applied Polarizability and C₄ Terms, Respectively^a

OPC		computed ΔG								
		AMBER					12–6–4 LJ	12–6 LJ	PLUMED	pol ($\text{\AA}^3$)
ion	exp ΔG^{50-52}	Rep #1	Rep #2	Rep #3	av $\pm$ std					
Ni(II)	−1.95	−1.73	−1.70	−1.74	$−1.72 \pm 0.02$	−10.54	−9.27	−2.10	−0.169	−24.81
Mg(II)	−1.33	−1.60	−1.67	−1.75	$−1.67 \pm 0.08$	−4.38	−7.68	−1.44	−1.00	−87.95
Cu(II)	−3.01	−2.91	−3.39	−3.17	$−3.16 \pm 0.24$	0.10	−8.07	−2.60	0.894	180.16
Zn(II)	−2.16	−2.10	−2.34	−1.81	$−2.08 \pm 0.27$	−6.45	−8.50	−2.08	0.432	67.31
Co(II)	−1.88	−1.77	−2.17	−1.64	$−1.86 \pm 0.28$	−4.93	−8.64	−2.30	0.357	50.43
Cu(I)	−2.78	−2.75	−3.29	−2.92	$−2.99 \pm 0.28$	−1.81	−2.03	−2.72	1.469	16.28
Fe(II)	−1.91	−1.84	−2.34	−1.58	$−1.92 \pm 0.39$	−4.77	−8.43	−2.19	0.369	39.35
Mn(II)	−1.91	−1.84	−1.65	−1.96	$−1.82 \pm 0.16$	−2.50	−6.79	−1.82	0.469	56.84
Cd(II)	−2.63	−2.78	−2.76	−2.80	$−2.78 \pm 0.02$	−0.23	−5.81	−2.65	0.819	124.21
Ca(II)	−1.27	−1.24	−1.22	−1.16	$−1.21 \pm 0.04$	−2.11	−3.84	−1.44	0.419	24.95
Ag(I)	−1.00	−0.95	−0.74	−1.00	$−0.90 \pm 0.14$	0.97	0.34	−1.11	1.650	94.84

^aThe free energy values are in kcal/mol.

incorporation of additional sites on the water molecule. This improvement includes a stronger dipole and quadrupole, enabling water molecules to form stronger interactions with other compounds than the weaker dipole and quadrupole

Table 5. Preferred Binding Mode for Each Metal Ion in the Three Water Models; Mono(%) and Bi(%) Indicate the Percentage of Monodentate and Bidentate Binding Modes in the Acetate–Metal Ion Complex^a

ion	acetate binding fashion					
	TIP3P		SPC/E		OPC	
Ni(II)	monodentate		monodentate		monodentate	
	Mono(99.98%)	Bi(0.02%)	Mono(100.00%)	Bi(0.00%)	Mono(100.00%)	Bi(0.00%)
Mg(II)	monodentate		monodentate		monodentate	
	Mono(99.95%)	Bi(0.05%)	Mono(100.00%)	Bi(0.00%)	Mono(100.00%)	Bi(0.00%)
Cu(II)	bidentate		bidentate		monodentate	
	Mono(30.96%)	Bi(69.04%)	Mono(31.79%)	Bi(68.21%)	Mono(99.81%)	Bi(0.19%)
Zn(II)	monodentate		monodentate		monodentate	
	Mono(98.47%)	Bi(1.53%)	Mono(99.59%)	Bi(0.41%)	Mono(100.00%)	Bi(0.00%)
Co(II)	monodentate		monodentate		monodentate	
	Mono(97.71%)	Bi(2.29%)	Mono(99.34%)	Bi(0.66%)	Mono(100.00%)	Bi(0.00%)
Cu(I)	monodentate		monodentate		monodentate	
	Mono(96.40%)	Bi(3.60%)	Mono(96.76%)	Bi(3.24%)	Mono(98.38%)	Bi(1.62%)
Fe(II)	monodentate		monodentate		monodentate	
	Mono(98.73%)	Bi(1.27%)	Mono(99.82%)	Bi(0.18%)	Mono(100.00%)	Bi(0.00%)
Mn(II)	monodentate		monodentate		monodentate	
	Mono(64.62%)	Bi(35.38%)	Mono(53.89%)	Bi(46.11%)	Mono(88.96%)	Bi(11.04%)
Cd(II)	bidentate		bidentate		bidentate	
	Mono(14.49%)	Bi(85.51%)	Mono(13.16%)	Bi(86.84%)	Mono(14.29%)	Bi(85.71%)
Ca(II)	bidentate		bidentate		bidentate	
	Mono(48.07%)	Bi(51.93%)	Mono(28.82%)	Bi(71.18%)	Mono(41.08%)	Bi(58.92%)
Ag(I)	bidentate		bidentate		bidentate	
	Mono(24.37%)	Bi(75.63%)	Mono(30.43%)	Bi(69.57%)	Mono(29.20%)	Bi(70.80%)

^aThe percentages were calculated by using the PMF profile for each acetate–metal ion complex and applying the Boltzmann distribution based on the minimum energy of each binding mode state.

present in the three-point water models.²⁶ The heightened polarization effects of the OPC water model can significantly impact the prediction of binding free energies by influencing hydrogen bonding, dipole–dipole interactions, and even van der Waals forces. Therefore, when utilizing the OPC in systems where water molecules play a pivotal role in mediating interactions, we have found that lower polarizability values for the chelator atoms become essential to accurately replicate experimental values.

3.2. Acetate Binding Mode in Different Water Models. When studying metal–acetate systems, we have two binding modes that could play a role in the coordination of the metal ion by acetate. In the monodentate binding mode, only one of the carboxylate oxygen atoms coordinate to the metal ion, while in the bidentate mode both carboxylate oxygen atoms coordinate the metal ion. Importantly, the binding mode of metal ions in a complex with carboxylate or acetate varies depending on the specific ligands involved and the binding site composition. The structures obtained/predicted by X-ray crystallography, FT-IR spectroscopy, and UV–vis spectroscopy show that acetate ligands are bonded to Zn(II) in a monodentate binding mode.⁵⁴ Structural studies on zinc-containing proteins have also revealed that the carboxyl group coordinates with zinc in bidentate, monodentate, and even intermediate between monodentate and bidentate fashions.⁵⁵ However, density functional theory (DFT) calculations indicate that the preference between monodentate and bidentate modes for Zn(II) depends on the properties of the binding site coordination environment. For instance, in a four-coordinate complex that is net positively charged, both mono- and bidentate coordination of carboxylates have similar energy. However, when additional ligands with negative

charges are introduced, the preference shifts toward the monodentate binding mode.⁵⁵

Previous studies show that both Ca(II) and Mg(II) prefer to coordinate with carboxylates in a monodentate binding fashion.⁵⁶ In contrast, bidentate binding becomes the preferred option when the attraction between the metal cation and the ligand's second oxygen is stronger than the interactions with nearby ligands.⁵⁶ In a combined experimental and computational approach it was suggested that Zn(II)–acetate binding likely involves both monodentate and bidentate geometries, while Mg(II)–acetate is predominantly in a monodentate binding mode.⁵⁷

A survey of Protein Data Bank (PDB) structures^{58,59} shows that Mg(II) binding sites have more monodentate carboxylates stabilized by coordinating water molecules while Ca(II) sites have more bidentate carboxylates stabilized by backbone interactions. In other words, water molecules stabilize monodentate binding in Mg(II) complexes, while backbone groups destabilize monodentate binding in Ca(II) complexes.⁵⁹ An X-ray crystallography study of different cobalt and nickel complexes in the presence of metal ion ligands, including tetraazamacrocycles, shows that smaller metal ions such as Co(III) bind to acetate in a bidentate mode, while larger metal ions such as Co(II) and Ni(II) in cross-bridged ligands tend to coordinate in a monodentate fashion. However, acetate prefers to have a bidentate binding mode with Co(II) and Ni(II) in the presence of unbridged ligands.⁶⁰

Overall, both experimental and quantum mechanical (QM) studies consistently support that acetate and carboxylate ligands do not exhibit a unique monolithic behavior when binding to metal ions.^{54–65} Various factors, including the coordination number, type of metal ion, its valence, charge, size, and the composition of the binding site, can impact the

tendency toward either bidentate or monodentate binding.^{56,61,62} Despite that, an examination of structures in the PDB database and the Cambridge Structure Database (CSD) reveals that the carboxylate group more commonly binds to various metal ions through a monodentate binding fashion in both small molecules and metalloproteins.^{56,58,59,66–69}

The type of water model can impact both the calculated binding free energy and the binding mode of the acetate molecule in MD simulations. This impact arises from a combination of factors, including the electron configuration of the metal ion, its ionic radius, and the characteristics of the water models employed in the simulations. The properties of the selected water model can influence interactions between ions and solvents as well as between ions and ligands. The variation in these interactions is why different polarizability values are needed to reproduce the experimental binding free energy in the OPC, TIP3P, and SPC/E water models.

Our findings demonstrate that the behavior of acetate, particularly in terms of monodentate and bidentate coordination, depends on both the nature of the metal ion and the specific water model used (Table 5).

We find that certain metal ions, including Cd(II) and Ca(II), exhibit consistent behavior, displaying a bidentate binding mode regardless of the water model employed. In TIP3P, the calcium ion exhibits a bidentate binding mode, but the monodentate form is nearly isoenergetic. While in SPC/E and OPC, it displays a bidentate preference with a slightly higher lying monodentate form (monodentate $\sim +0.75$ kcal/mol higher). On the other hand, Cd(II) demonstrates a bidentate (monodentate $\sim +1.5$ kcal/mol higher) mode consistently across all water models.

Figure 1 shows the detailed free energy profile of the Cd(II)–acetate complex defined by the interaction between

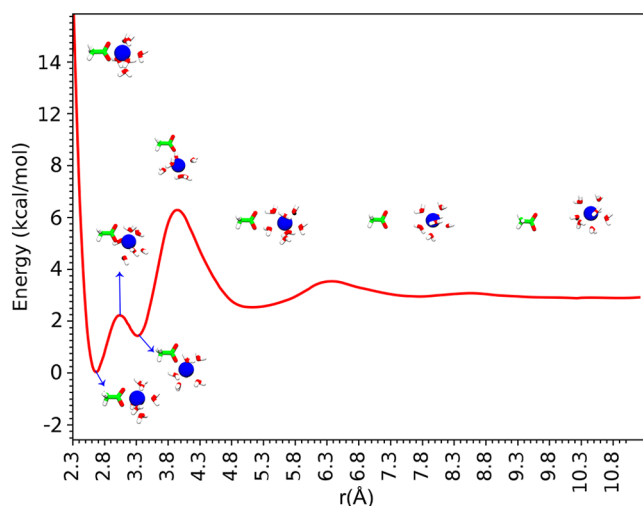


Figure 1. Depiction of the PMF energy profile of the Cd(II) and acetate complex in the TIP3P water model, accompanied by snapshots at various points along the profile.

the acetate oxygen atoms and the metal ion using the 12–6–4 LJ nonbonded model. An examination of PDB structures uncovers that Cd(II) participates in both bidentate and monodentate binding modes with the carboxylate groups of metalloproteins.⁵⁸ Our results demonstrate that this metal ion displays a preference of ~ 1.5 kcal/mol for bidentate binding with the acetate molecule.

Cu(II) exhibits different behaviors depending on the water model used. In three-point water models (TIP3P and SPC/E) it prefers a bidentate interaction (with a distinct monodentate minimum ~ 0.1 – 0.5 kcal/mol higher in energy), while in OPC (4-point) water it prefers monodentate binding and there is only a slight shoulder at the location where the bidentate complex would appear. The reason for this is not entirely clear, but one can hypothesize that differential water–water interactions around the ions may play a role. This is discussed further below.

3.2.1. Acetate Binding Mode in Water Models with Three Interaction Sites. Overall, we noticed consistent binding behavior of the acetate molecule in both the TIP3P and SPC/E water models (Table 5). This outcome was not surprising because both water models are characterized by three-point configurations with nearly identical O–H bond lengths and H–O–H angles.¹⁰ Both predict the acetate molecule to prefer to coordinate with Ni(II), Mg(II), Zn(II), Co(II), Fe(II), and Mn(II) ions in a monodentate fashion (Figure 2).

Previous experimental studies have demonstrated that the binding mode of the carboxylic group with Ag(I) can vary from bidentate to monodentate depending on the ligands.^{70,71} Despite that our results predict that acetate shows an ~ 0.75 kcal/mol preference for bidentate binding mode with Ag(I), featuring a shoulder at the monodentate position (Figure 2). Cu(I) prefers monodentate coordination but has a clear shoulder for the bidentate coordination mode (monodentate (bidentate ~ 1.5 kcal/mol higher in energy shoulder)). Overall, we find that most divalent metal ions, including Ni(II), Mg(II), Zn(II), Co(II), Fe(II), and Mn(II) as well as Cu(I), tend to adopt a monodentate binding configuration with the ligand in all three water models, exhibiting a preference of approximately 1.5–3 kcal/mol (~ 0.3 and ~ 0.1 kcal/mol preference for Mn(II) in TIP3P and SPC/E, respectively). This trend aligns with the findings from statistical analyses conducted on structures from the CSD and PDB databases.^{56,58,59,66–69} However, for Ag(I), Ca(II), and Cd(II) the bidentate binding mode is preferred in both the TIP3P and SPC/E water models. While Cu(II) in these water models prefers the bidentate binding mode with acetate (Figures 2 and 3), in the OPC water model this metal ion shows a preference for monodentate binding (bidentate shoulder ~ 3 kcal/mol higher in energy).

Experimental studies indicate that copper can establish diverse complexes with carboxylate ligands, commonly serving as bidentate ligand.⁶⁵ However, there has also been the observation of an ongoing shift between monodentate and bidentate bonding modes in carboxylate.⁶⁴ Additionally, in the Cu(II) complex with Hmbm (1-methyl-1H-benzo[d]imidazol-2-yl)methanol, acetate coordinates with the metal ion in a monodentate fashion.⁵⁴

In fact, among divalent metal ions in the TIP3P and SPC/E water models, the size of the ion plays an important role in their binding mode with acetate. In comparison to other metal ions, specifically Ca(II) and Cd(II), the larger radii of these ions result in bidentate coordination, facilitated by the available space for engaging with multiple oxygen atoms from the acetate. Conversely, the smaller ions, excluding Cu(II), exhibit a higher degree of selectivity, potentially favoring monodentate binding configurations (Table 5).

3.2.2. Acetate Binding Mode in Water Models with Four Interaction Sites. In the OPC water model, acetate binds to the majority of metal ions, including Ni(II), Mg(II), Cu(II),

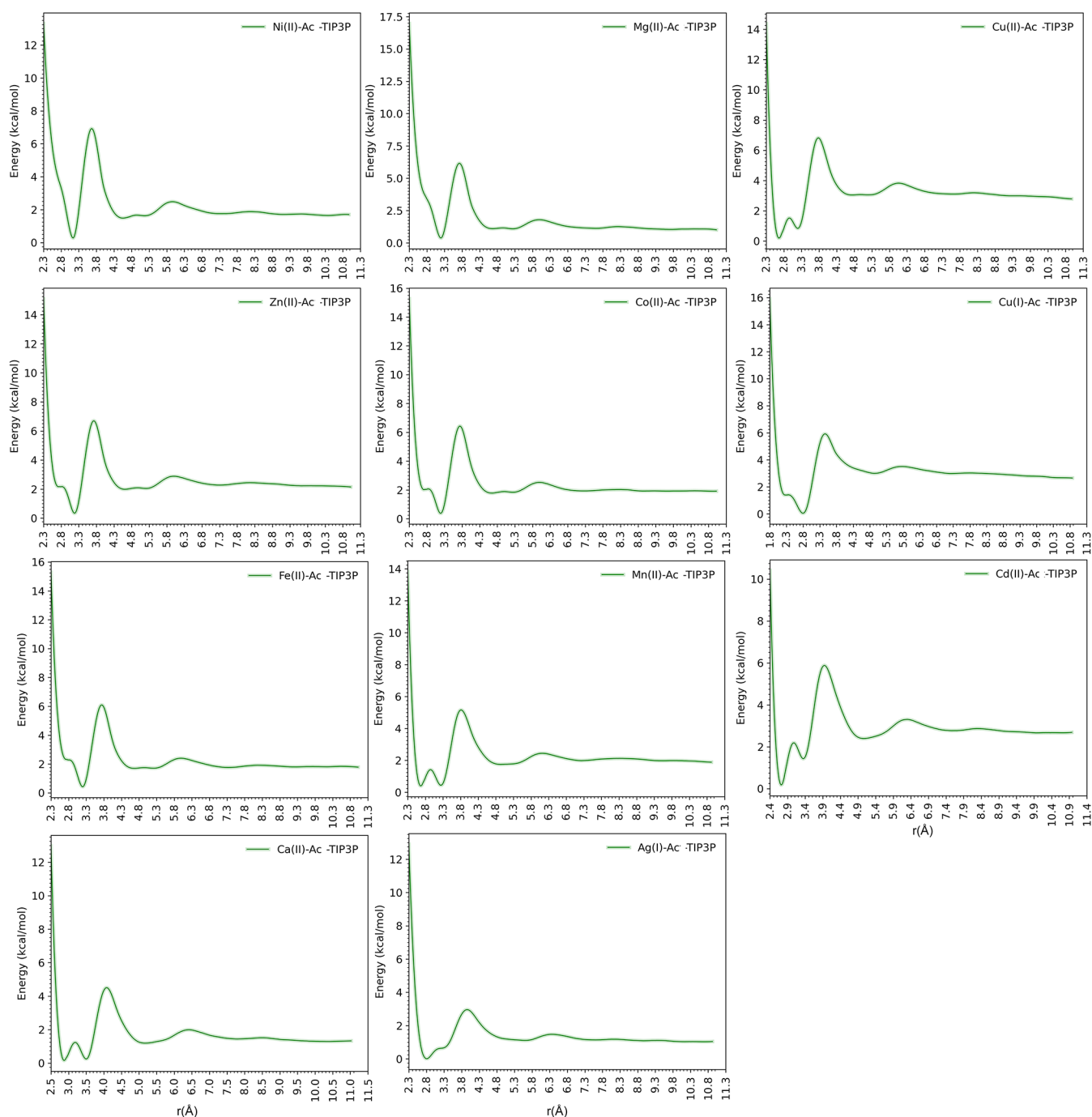


Figure 2. PMF free energy profiles of metal ion–acetate complexes in the TIP3P water model. Ac stands for the acetate molecule. The first local minimum, occurring at approximately 2.8 Å (2.3 Å for Cu(I)), corresponds to the bidentate binding mode. The second local minimum, observed at around 3–3.5 Å (2.8 Å for Cu(I)), shows the monodentate binding mode.

Zn(II), Co(II), Fe(II), Mn(II), and Cu(I) through monodentate coordination, showing a preference of approximately 1–1.5 kcal/mol for monodentate in the case of Cu(II), Mn(II), and Cu(I) over the bidentate coordination mode (Figure 4).

As discussed, statistical analyses of structures from both the PDB and the CSD databases consistently show that most metal ions often favor a monodentate binding mode with carboxyl groups. Remarkably, when using the OPC water model, which more accurately mimics liquid water properties, the results are closer to the experimental data, suggesting a tendency toward

monodentate acetate binding with metal ions. As noted previously the binding behavior of metal ions depends on various factors.^{48,49}

As shown in Table 5 and Figures 2–4, acetate exhibits a change in behavior when it forms complexes with Cu(II). This change is evident as the ligand alters its coordination mode from bidentate to monodentate. In the case of Zn(II), Co(II), and Fe(II), the shoulder at the bidentate position disappears. This suggests that in the presence of OPC water model, these metal ions shift from their initial monodentate (bidentate ~ shoulder 2–3 kcal/mol higher in energy) binding mode to a

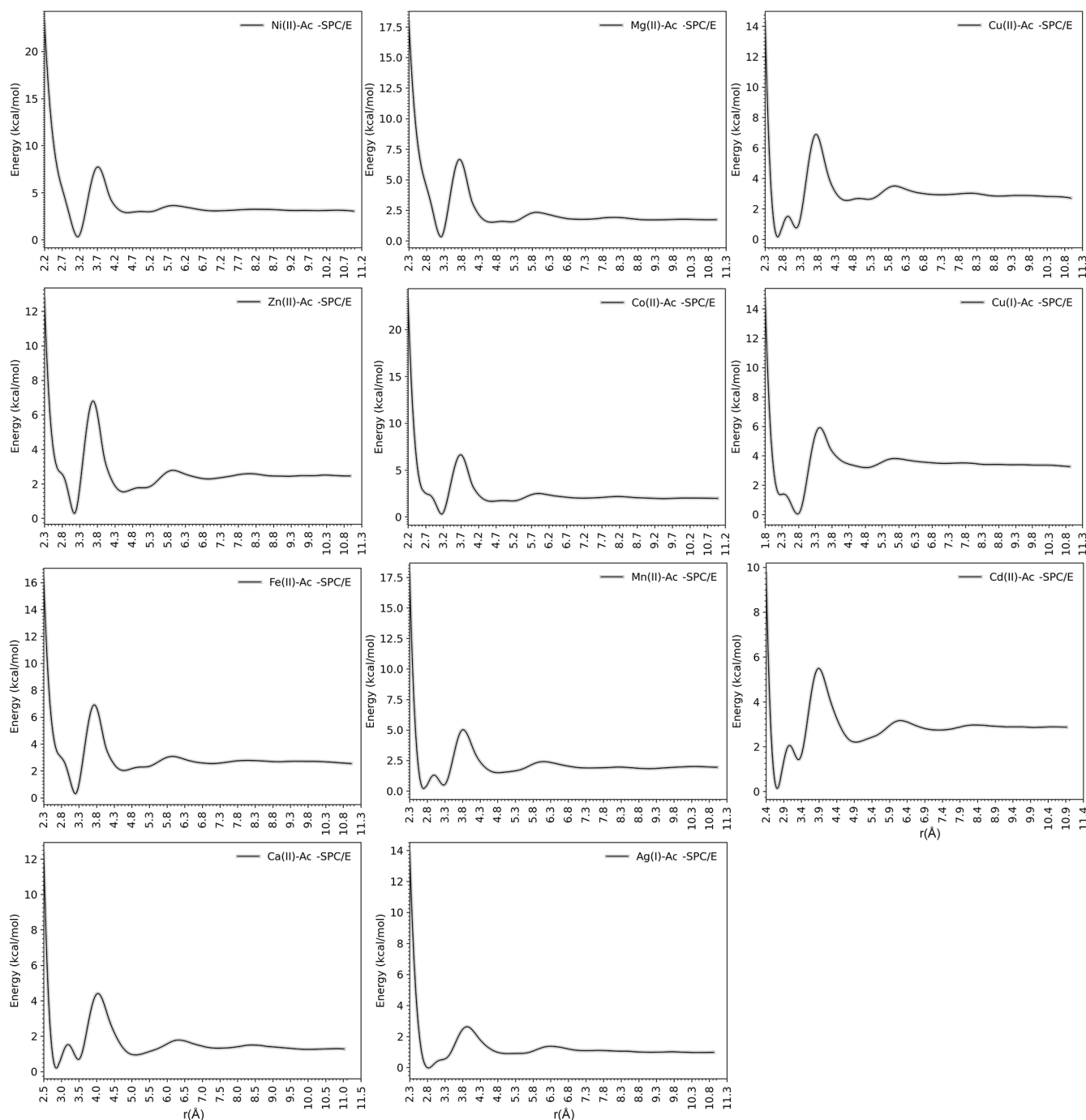


Figure 3. PMF free energy profiles of metal ion–acetate complexes in the SPC/E water model. Ac stands for the acetate molecule. The first shoulder, at approximately 2.8 Å (2.3 Å for Cu(I)), shows the presence of a bidentate binding mode. The second shoulder, located at approximately 3–3.5 Å (2.8 Å for Cu(I)), indicates the occurrence of the monodentate binding mode.

monodentate binding mode. Although carboxyl groups typically exhibit monodentate coordination with metal ions,^{56,58,59,66–69} it is important to note that there is no consistent trend in the binding modes of carboxyl groups with metal ions. Both bidentate and monodentate coordination have been observed in various acetate–metal ion complexes.^{54–65} and no experimental data are available for the preferences of a single acetate coordinating with metal ions in aqueous solution.

3.3. Binding Free Energy and the Chelator Atoms Polarizability. The polarizability of the chelator atom plays a

pivotal role in accurately reproducing the experimental binding energy. This parameter is influenced by various factors, including the ion charge, the number of unpaired electrons in the ion's outermost subshell, solvent effects, and ligand structure. In our study, we obtained nearly identical polarizability values for the closely related water models TIP3P and SPC/E. Overall, there was a slight increase in the applied polarizability for the chelating atom within the SPC/E model compared to that of the TIP3P water model. Significantly, in order to effectively replicate the experimental binding energy between metal ions and acetate, it was necessary to employ a

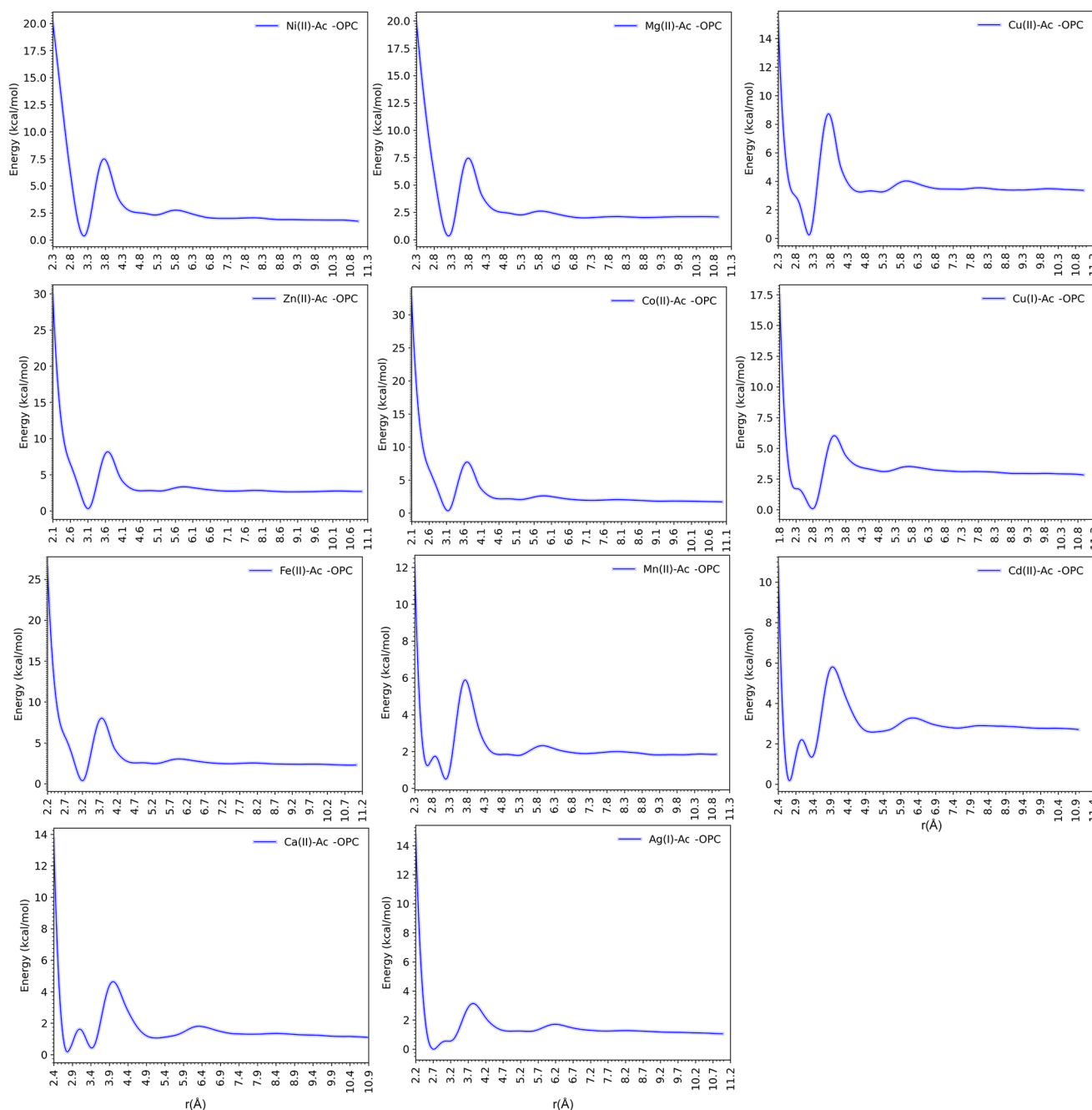


Figure 4. PMF free energy profiles of the metal ion–acetate complexes in the OPC water model. Ac stands for the acetate molecule. The first local minimum, found at around 2.8 Å (2.3 Å for Cu(I)), corresponds to the bidentate binding mode. The second local minimum, detected at approximately 3–3.5 Å (2.8 Å for Cu(I)), indicates the presence of the monodentate binding mode.

reduced polarizability value for the ligand chelator atoms in the OPC water model.

When considering alkaline earth metal ions, it becomes evident that ions with larger radii require higher polarizability for the chelator atom to accurately reproduce the experimental binding energy within all three water models (see Tables 2 and 4). Another significant implication is observed for Ca(II) and Mg(II), where the chelator atom with a bidentate binding mode needs greater polarizability. To observe this effect, one can compare the Ca(II) and Mg(II) PMF energy profiles presented in Figures 2–4 with the corresponding polarizability values listed in Tables 2–4. This trend in ionic radius holds true for ion families sharing similar electron configuration, such

as Zn(II) and Cd(II), Cu(I) and Ag(I), and Mg(II) and Ca(II). Essentially, ions with larger radii lead to higher polarizability on the chelator atoms. In a previous work, this behavior was observed for the nitrogen atom of imidazole.¹⁰ The polarizability of the chelator atom in the acetate ligand was found to be directly proportional to the radius of the ions within the same family. This relationship was applicable to divalent metal ions in three-point water models. In the case of OPC water, this correlation was evident only for monovalent metal ions, implying that larger monovalent metal ions required higher polarizability. However, in systems with three interaction site water models, a lower polarizability on the oxygen atoms was required with larger metal ions to

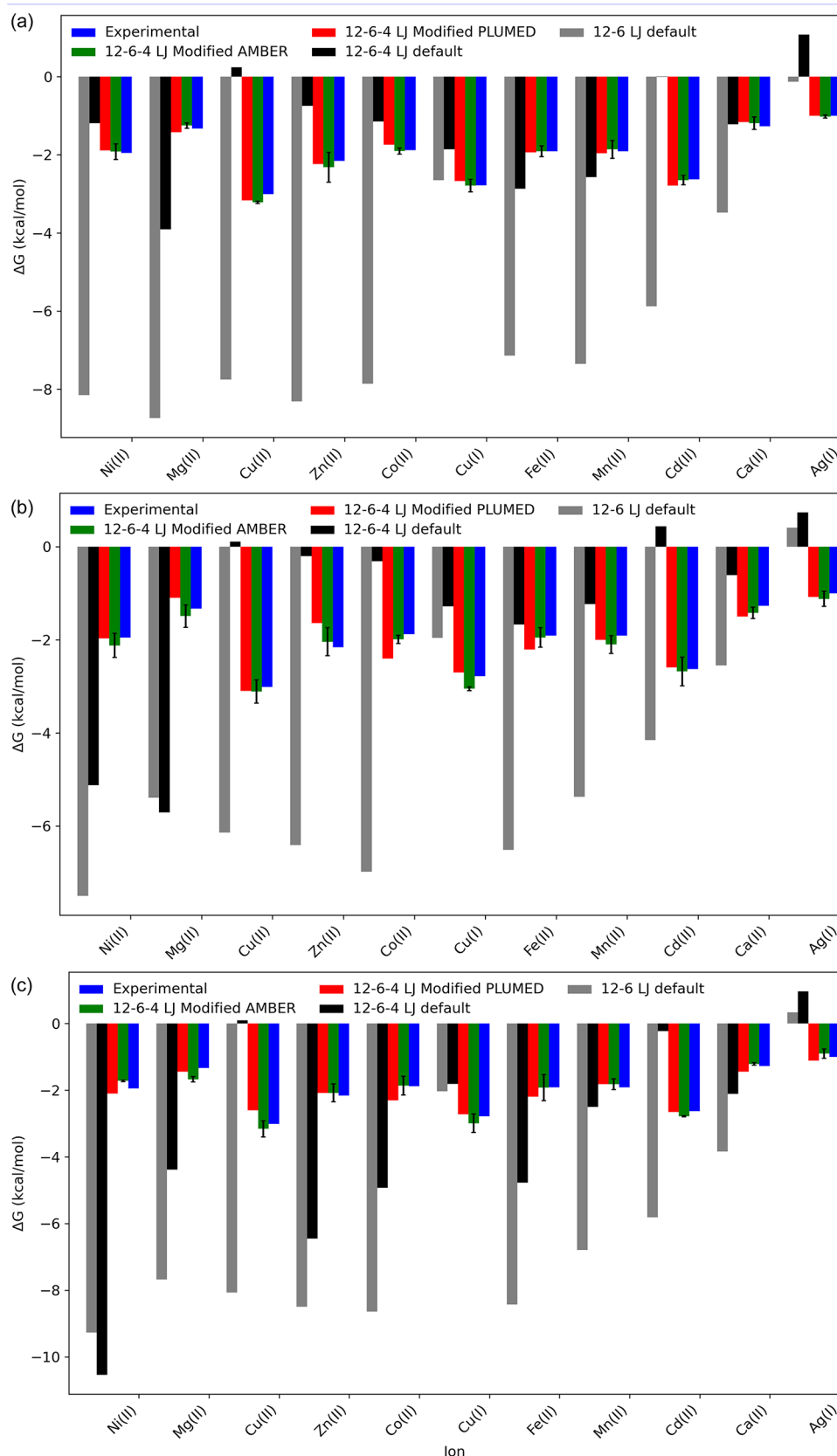


Figure 5. Obtained binding free energies using the default 12–6 LJ nonbonded model (gray bars), the default 12–6–4 LJ nonbonded model (black bars), an optimized 12–6–4 LJ model as obtained using AMBER (green bars, capped lines indicate standard deviations), and a modified 12–6–4 LJ model as obtained using PLUMED (red bars). The blue bars show the experimental binding free energy. (a, b, c) Results obtained using the TIP3P, SPC/E, and OPC water models, respectively.

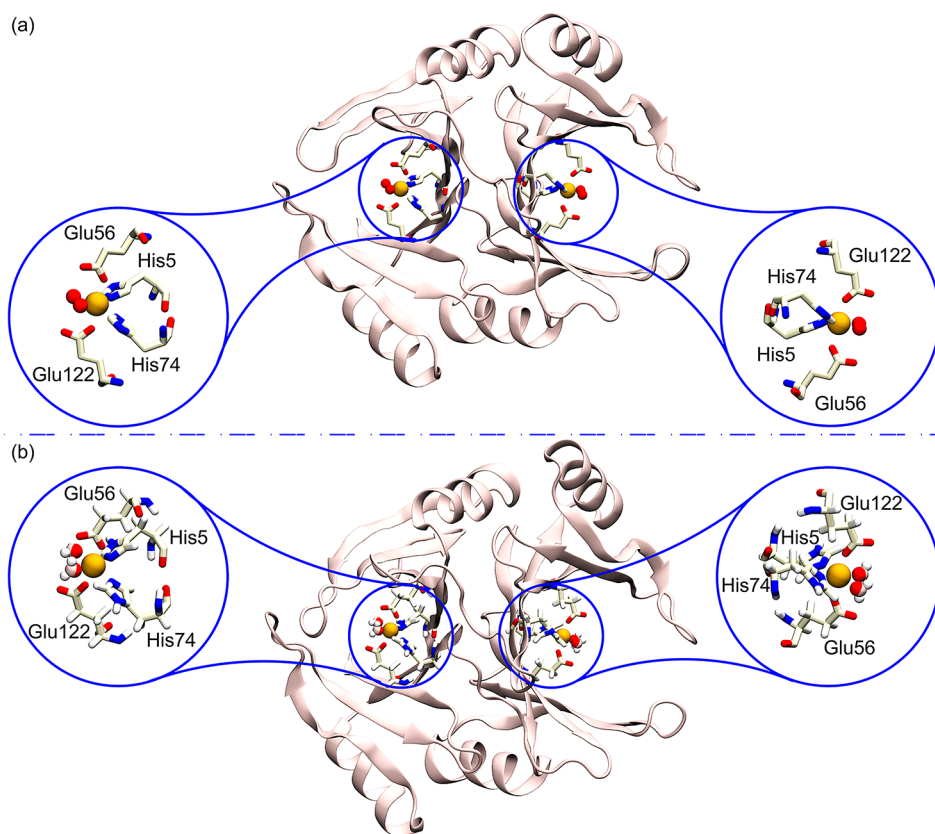


Figure 6. (a) shows the crystal structure of the Glx 1 protein, with the residues (pale yellow) coordinating with nickel (gold-colored balls). (b) displays the last snapshot of the MD simulations, illustrating the residues coordinating with the metal ion. As depicted in (b), the metal ion coordination is maintained using our optimized 12–6–4 LJ parameter set after 200 ns of MD simulations. The protein structure, water molecules (panel (a))/OPC water molecules (panel (b)) in the metal binding site, and the binding site residues are depicted as cartoon, VDW, and licorice models.

replicate the experimental binding free energy (as observed in Cu(I) and Ag(I)).

In the case of monovalent metal ions, the behavior of the chelating atoms in acetate mirrors that of the chelating atoms in the imidazole molecule used in previous studies.¹⁰ This observation indicates that in systems involving monovalent metal ions the current standard 12–6 LJ and 12–6–4 LJ nonbonded models generally underestimate induced dipole interactions. As a result, chelator atoms require higher polarizability to effectively reproduce the experimental binding free energy compared to systems with divalent metal ions. This holds true for divalent metal ions such as Cu(II) and Cd(II) in both the TIP3P and SPC/E water models, which are involved in bidentate coordination with acetate.

In the context of the smaller ions, such as Ni(II) and Mg(II) in the OPC water model, we found it necessary to employ negative polarizability values to accurately reproduce the experimental binding free energy (see Table 4). This phenomenon could be attributed to two potential reasons. First, with regards to the nature of the metal ions, the smaller ones have a higher charge density, leading to a more tightly bound electron cloud around them. Accordingly, they tend to form stronger interactions with the chelator atoms. Second, in terms of the water model OPC water models are optimized in terms of dipole and quadrupole moments to provide a more precise representation of liquid water's dipole moments and polarization behavior. Thus, we speculate that the utilization of

negative polarizability values became necessary in order to correct the overestimation observed in the 12–6 LJ and standard 12–6–4 LJ models for these smaller ions.

3.4. Comparison of 12–6 LJ and 12–6–4 LJ Models with the Modified 12–6–4 LJ Model. Figure 5 compares the experimental and calculated binding energies of 11 metal ions. Various parameter sets were employed to assess the binding energies of 11 different metal ions. Specifically, binding energies were computed utilizing three distinct models: the default 12–6 LJ model, the default 12–6–4 LJ model, and the modified 12–6–4 LJ model, extracted from AMBER PMF studies. Furthermore, to enhance the robustness of the findings, the binding energies associated with the modified 12–6–4 LJ model were additionally validated using PLUMED PMF simulations.

The standard 12–6 LJ model underestimates the binding interactions with Ag(I). However, it seems that the interactions of Cu(I) with the acetate molecule are more favorable. This is because the Cu(I) ion binds to acetate in monodentate binding mode with a shoulder at the bidentate position, allowing the ligand to have a greater space for establishing multiple interactions with the metal ion. For the other metal ions studied, this model tends to overestimate the binding interactions of acetate, regardless of the water model employed (indicated by the gray bars in Figure 5). Another significant implication arises when using the standard 12–6 LJ

nonbonded model. That is as the radius of the metal ion increases, the degree of overestimation generally decreases.

Incorporating the default C4 terms into the 12–6–4 LJ model (the 12–6–4 LJ nonbonded model) causes a significant adjustment in the binding free energy for most metal ions with acetate. However, the resulting free energy values deviate from the experimental binding free energies (shown by the black bars in Figure 5).

Compared to the OPC water model, the default 12–6–4 LJ parameters underestimate the binding free energy of the majority of metal ions in water models with three interaction sites. However, most of the metal ions exhibit stronger interactions with acetate when the default 12–6–4 LJ parameters are used in conjunction with the OPC water model.

By implementing the modified C4 terms in this study within the newly developed 12–6–4 LJ nonbonded model, we successfully reproduced the experimental binding free energies of metal ions with acetate, achieving an acceptable margin of error of approximately 0.35 kcal/mol (depicted as green bars in Figure 5). Our findings from the AMBER PMF simulations align closely with those obtained using PLUMED as an external validation to compute the PMF, with a margin of error of around 0.5 kcal/mol.

To assess our parameters beyond a system composed exclusively of acetate and a metal ion, we applied them to a system containing the *Escherichia coli* Glyoxalase I metalloprotein (PDB ID: 1F9Z).³⁵ A previous study on this protein indicates that the z12–6 parameters fail to maintain the coordination modes of both Ni(II) binding sites after 80 ns of MD simulations.⁷² This metalloprotein is a homodimer comprising two metal-binding sites, with each binding site including His5, His74, Glu122, and Glu56 and two water molecules coordinating with one nickel (see Figure 6).

The results from this study⁷² show that two histidine residues contributing to the metal coordination sites lose their interactions with the metal ion. Two oxygen atoms from the negatively charged residues in the binding site then take over and coordinate with the metal ion in a bidentate mode.

Our results demonstrate that after a 200 ns MD simulation using our optimized 12–6–4 parameter set for both negatively charged residues (current work) and histidine,¹⁰ the two histidine residues in both metal binding sites maintain their interactions with the metal ions (Figure 6). Meanwhile, the negatively charged residues (GLU56 and GLU122) coordinate with the metal ion in a monodentate mode, consistent with the crystal structure of the metalloprotein. In the crystal structure, two water molecules coordinate with the metal ions, as Figure 6b shows the two water molecules remain in both Ni(II) binding sites of the metalloprotein during the simulation. This outcome confirms the transferability of our optimized 12–6–4 LJ parameters to other systems containing proteins.

4. CONCLUSIONS

In this work, we optimized the 12–6–4 LJ nonbonded model to accurately reproduce the experimental binding free energy between acetate and different metal ions. These parameters can be further extended to metalloproteins and metal transporters, facilitating the successful prediction of metal ion interactions with acidic residues (aspartate and glutamate). As previously demonstrated,⁸ the 12–6–4 LJ parameters enhance the metal ion interactions in aqueous solutions via the inclusion of dipole-induced interactions between the metal ions and water. However, our findings in this study reveal that while the

standard 12–6–4 LJ parameters improve metal ion interactions with acetate, they still deviate from the experimental binding free energy values. This work effectively addresses this issue by precisely reproducing the experimental binding free energy of various metal ions with acetate by parametrizing the 12–6–4 model specifically for acetate. Our investigation demonstrates that optimizing the experimental binding free energy across different solvent models depends on several factors, including the metal ion's nature (e.g., the count of unpaired electrons in its outermost subshell), the solvent's characteristics, and the ligand. Successfully accounting for these factors can be achieved by adjusting the C4 terms. The obtained parameters are broadly applicable in a large number of systems involving metal ions.

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Notes

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